



Activity Overview

Recently, you've been thinking about identifying good data sources that would be useful for analysis. You also spent some time in a previous activity exploring a public dataset in BigQuery and writing some basic SQL queries. In addition to using public data in BigQuery, your future data career will involve importing data from other sources. In this activity, you will create a custom table and dataset that you'll load into a new table and query.

By the time you complete this activity, you will be able to load your own data into BigQuery for analysis. This will enable you to import your own data sources into BigQuery, which is a skill that will enable you to more effectively analyze data from different sources.



Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

To get started, download the baby names data zip file. This file contains about 7 MB of data about popular baby names from the U.S. Social Security Administration website.

Select the link to the baby names data zip file to download it.

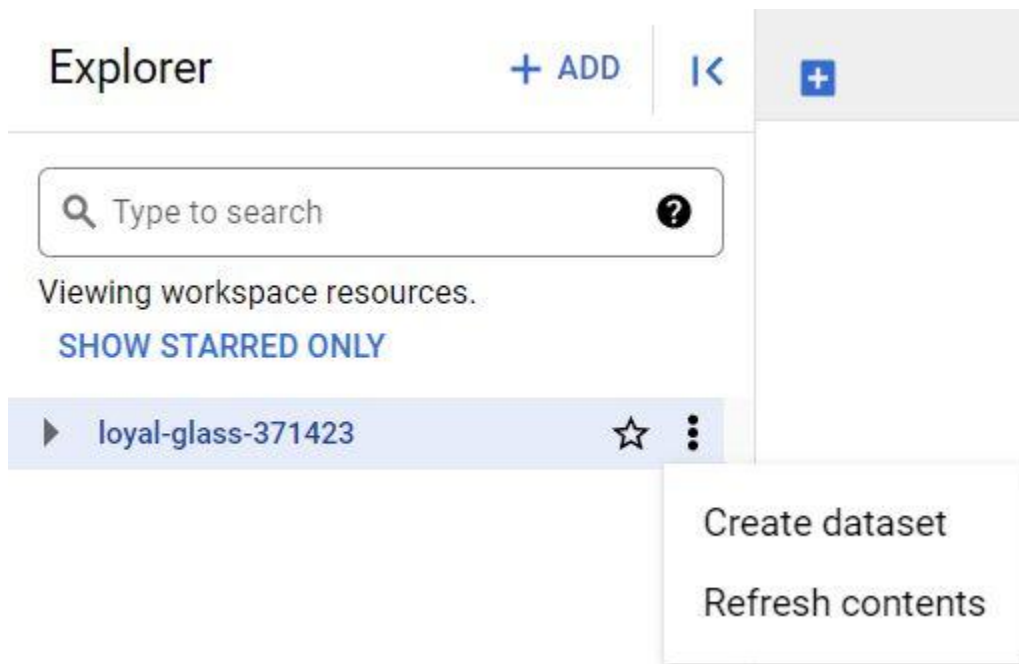
Link to baby names data: [names.zip](#)

Unzip the file you downloaded onto your computer to access it on BigQuery. Once you have unzipped the file, find a `.pdf` file titled `NationalReadMe` that contains more information about the dataset. This dataset tracks the popularity of baby names for each year; you can find text files labeled by the year they contain. Open `yob2014.txt` to preview the data. You will notice that it's a `.csv` file with three columns. Remember where you saved this folder so you can reference it later.

Before uploading your `.txt` file and creating a table to query, you will need to create a dataset to upload your data into and store your tables.

1. From the BigQuery console, go to the **Explorer** pane in your workspace and **select** the three dots next to your project to open a menu. From here, select **Create dataset**. Note that unless you have already specified your own project name, a

unique name is assigned to your project by BigQuery, typically in the format of two words and a number, separated by hyphens (e.g., `loyal-glass-371423` in the image below).



2. This will open the **Create dataset** menu. This is where you will fill out information about the dataset. Input the **Dataset ID** as `babynames` and set the **Data location** to **Multi-region (US)**. Once you have finished filling out this information, select the blue **CREATE DATASET** button at the bottom of the menu.

Create dataset

Dataset ID *

babynames

Letters, numbers, and underscores allowed

Location type ?

☐

Region

Specify a region to colocate your datasets with other GCP services.

☒

Multi-region

Allow BigQuery to select a region within a group to achieve higher quote limits.

Multi-region *

US (multiple regions in United States)



Default table expiration

☐

Enable table expiration ?

Default maximum table age

Days

Advanced options



Encryption ?

☒

Google-managed encryption key

No configuration required

☐

Customer-managed encryption key (CMEK)

Manage via [Google Cloud Key Management Service](#)

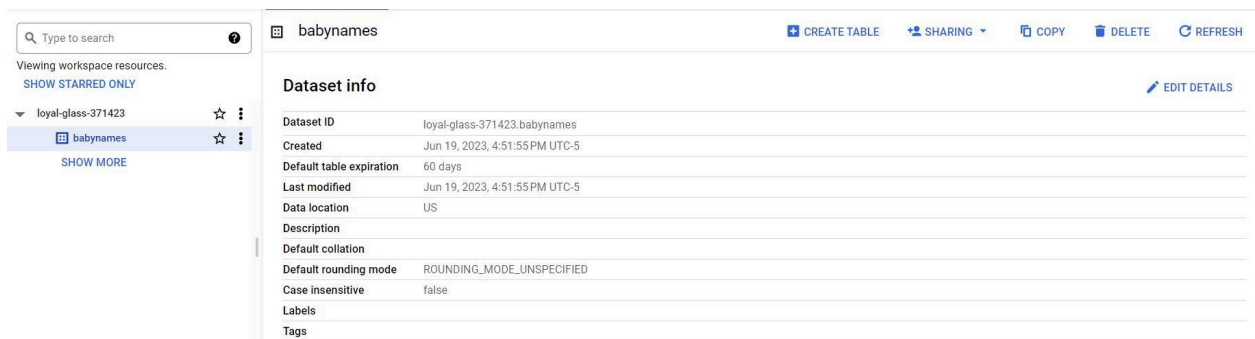
Case Insensitive

CREATE DATASET

CANCEL

Now that you have a custom dataset stored in your project space, this is where you will add your table.

1. Select the newly created **babynames** dataset. Check the tabs in your **Dataset info** window and select the first blue **+ CREATE TABLE** button. This will open another menu in your console.



2. In the **Source** section, select the **Upload option** in **Create table from**. Then select **Browse** to open your files. Find and open the **yob2014.txt** file. Set the file format to **.csv**. In the **Destination** section, in the **Table data** box, name your table as **names_2014**. For **Schema**, select **Edit as text** and input the following code:

```
name:string,
```

```
gender:string,
```

```
count: integer
```

3. This will establish the data types of the three columns in the table. Leave the other parameters as they are, and select **Create table**.

Create table

Source

Create table from:

Upload ▼

Select file: ?

yob2014.txt

Browse

File format:

CSV ▼

Destination

☒ Search for a project

☐ Enter a project name

Project name

test ▼

Dataset name

babynames ▼

Table type ?

Native table ▼

Table name

names_2014

Schema

Auto detect

☐ Schema and input parameters

☐ Edit as text

```
1 name:string,gender:string,count:integer|
```

Partition and cluster settings

Partitioning: ?

No partitioning ▼

Create table

Cancel

Once you have created your table titled `names_2014`, it will appear in your explorer pane under the dataset `babynames` that you created earlier.

Explorer

+ ADD



🔍 Type to search



Viewing workspace resources.

[SHOW STARRED ONLY](#)

▼	loyal-glass-371423	☆	⋮
▶	↻ External connections		⋮
▼	📊 babynames	☆	⋮
	📊 names_2014	☆	⋮

4. Select the table to open it in your workspace. Here, you can check the table schema. Then, go to the **Preview** tab to explore your data. The table should have three columns: `name`, `gender`, and `count`.

names_2014

QUERY

SHARE

COPY

REFRESH

SCHEMA

DETAILS

PREVIEW

LINEAGE

DATA PROFILE

DATA QUALITY

Row	name	gender	count
1	Emma	F	20941
2	Olivia	F	19817
3	Sophia	F	18628
4	Isabella	F	17102
5	Ava	F	15708
6	Mia	F	13516
7	Emily	F	12650
8	Abigail	F	12093

Now that your table is set up, you’re ready to start writing queries and answering questions about this data. For example, let’s say you were interested in the top five baby names for boys in the United States in 2014.

Select **COMPOSE NEW QUERY** to start a new query for this table. Then, enter this code:

SELECT

name,

```
count
```

```
FROM
```

```
your-project.babynames.names_2014
```

```
WHERE
```

```
gender = 'M'
```

```
ORDER BY
```

```
count DESC
```

```
LIMIT
```

NOTE: Making sure that your **FROM** statement is correct is key to making this query run! The dataset needs the query to tell it the location of the table you just uploaded so that it can fetch the data. It's like giving the query a map to your table. That map will include your unique BigQuery project name (replace **your-project** in the code above with your unique project name), the dataset name (**babynames**), and the table name (**names_2014**). The location names for each of these elements are separated by periods. The final result will be something like this:

```
loyal-glass-371423.babynames.names_2014
```

Note that **loyal-glass-371423** is just an example of a project name. You must use your project's name in your **FROM** statement.

This query selects the **name** and **count** columns from the **names_2014** table. Using the **WHERE** clause, you are filtering for a specific gender for your results. Then, you're sorting how you want your results to appear with **ORDER BY**. Because you are ordering by the **count** in descending order, you will get names and the corresponding count from largest to smallest. Finally, **LIMIT** tells SQL to only return the top five most popular names and their counts.

Once you have input this in your console, select **RUN** to get your query results.

If you are comfortable creating your own custom tables, try uploading more files from the baby names dataset into tables you can query. For example, you could upload each of the files from 2015 to 2019 to find the top baby names for those years.

1.

Question 1

Reflection

After running the query on your new table, what was the third-most popular baby name for boys in 2014?

Noah

Mason

Jacob

William

Correct

To find that Mason was the third most popular baby name for boys in 2014, you queried your custom table and checked the results. Going forward, you'll be able to upload your own data sources into BigQuery for future analysis projects. This will allow you to practice writing SQL queries for more data sources, which will be a key skill as a data analyst.

1 / 1 point

2.

Question 2

In this activity, you explored public data in BigQuery and used it to create a custom table. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- Why is being able to use data from different sources useful as a data analyst?
- How can you use BigQuery custom tables and datasets in your future analysis projects?

If I am able to use data from different sources, then it becomes very useful because I can analyze any data that I want to analyze, and in the real world data will be most vast, and for analysis we want different source data. If we can't analyze from different sources, then we can't find any insights, so it is very useful, and obviously all analysts must learn to use data from different sources. Yeah, of course, I will upload my data in my project to create data, and then I will upload my desired dataset, which I want to analyze. Then in the preview section we also can see the table. Then I use the query for my required insight. In BigQuery it's a very trusted and useful tool for analyzing the data.

Feedback

Congratulations! In this activity, you created a new dataset within your project, uploaded a .csv file to create a new table, and ran a SQL query. A good response would include that being able to evaluate and use different data sources allows you access more data.

As a data analyst, being able to evaluate data sources and use the appropriate tool to analyze them is important. For instance, you were able to use SQL to analyze a dataset that was previously stored on your computer as a .csv file.

